

Equipment Inspection Report

Report number	INS-2026-0847
Equipment tag	P-101-A
Equipment description	Centrifugal pump, crude charge, motor driven
Unit / area	CDU-1, Pump House B
Service	Crude oil, 62 °C, continuous
Inspection date	28 August 2026
Inspector	R. Nair, Inspection Engineer
Procedure reference	SOP-PMP-114 Rev. 4

1. Measured values

Parameter	Measured	Instrument
Casing wall thickness (minimum)	4.1 mm	UT gauge CM-114
Bearing vibration, drive end	7.8 mm/s RMS	Vib analyser VA-07
Bearing vibration, non-drive end	3.9 mm/s RMS	Vib analyser VA-07
Bearing housing temperature	74 °C	IR thermometer IR-22
Discharge pressure	18.2 bar, fluctuating $\pm$ 1.1 bar	PI-1014 / panel trend
Motor current	fluctuating $\pm$ 9 %	Panel trend
Suction strainer differential	0.41 bar	PDI-1009
Mechanical seal	Intermittent weeping observed	Visual

2. Observations

- 2.1 A pronounced rattling noise was audible at the pump during the inspection. The noise was described by the field operator as similar to gravel or marbles moving inside the casing. The noise was intermittent and coincided with periods of unstable discharge pressure.
- 2.2 Discharge pressure and motor current were both observed to fluctuate over a ten minute period at the panel. Flow rate on FI-1012 was also unsteady.
- 2.3 Minor corrosion pitting was noted on the suction nozzle flange face. No through-wall indication.
- 2.4 One of four coupling guard fasteners was missing. Guard remained secured by the remaining three.
- 2.5 Lubrication log last signed 12 July 2026. Quarterly lubrication verification is therefore overdue at the date of this inspection.
- 2.6 Suction vessel V-104 level was noted at 21 % during the inspection, which is at the low end of the normal operating band.

3. Inspector remarks

Several parameters appear to be outside the limits stated in the governing procedure. The combination of audible noise with unstable discharge pressure and unstable motor current is noted for engineering assessment. Recommend review before the next scheduled run.

This report records measured values and observations only. Disposition is to be determined by the Rotating Equipment Engineer in accordance with SOP-PMP-114 Section 5.